

Susanta Das, Ph.D.

Research Associate

Computational Life Sciences (CLS)
Cleveland Clinic Research, Cleveland Clinic Foundation
9500 Euclid Ave. NA2 #39, Cleveland, OH 44195, USA

✉ dass11@ccf.org
✉ susanta.chemistry@gmail.com
🌐 [isusanta.github.io](https://github.com/isusanta)
in [dr-das](#)

ORCID: 0000-0001-7981-5162
Google Scholar: 672 citations (August 2026)

Areas of Expertise

Quantum Computing: Heterogeneous quantum–classical computing; Sample-based Quantum Diagonalization (SQD) and extended SQD (ext-SQD); Embedded Wavefunction (EWF) fragmentation; IBM Quantum Platform (QPU execution, active orbital fragments); Qiskit; reduced density matrix reconstruction; democratic energy partitioning; quantum hardware benchmarking for molecular energetics.

Electronic Structure: High-level correlated wavefunction methods: Coupled-Cluster (CCSD, CCSD(T)), Full Configuration Interaction (FCI), Møller–Plesset perturbation theory; multi-reference electron correlation; Density Functional Theory (DFT); *Ab initio* Molecular Dynamics (AIMD); Conceptual DFT; Time-Dependent DFT.

Fusion/Ionic Materials: Heterogeneous quantum–classical electronic structure of FLiBe (2LiF–BeF₂) molten salt, a leading fusion-reactor blanket material: clusters from *ab initio* MD partitioned into atom-centered EWF fragments, largest fragments solved on IBM quantum hardware via extended sample-based quantum diagonalization (ext-SQD). Tritium speciation and binding energetics; fragment ground-state energies within 0.7 kcal/mol of full CI (0.3 kcal/mol mean absolute deviation across nine clusters); first quantum-hardware demonstration for a charged ionic / inorganic molten-salt system.

QM/MM Methods: Hybrid Quantum Mechanical / Molecular Mechanical (QM/MM) simulation; CHARMM-based QM/MM (CHARMM–QCHEM interface); enzyme catalysis; biomacromolecule dynamics; EnzyDock docking methodology development (simulated annealing MD and Monte Carlo sampling); QM region size convergence studies.

QM + ML Spectroscopy: QM/ML pipelines for NMR chemical shift prediction; Collisional Cross Section (CCS) prediction for metabolite annotation; AutoGraph autonomous graph-based conformational clustering (Louvain algorithm); web server development for metabolomics characterization (pomics.org); Snakemake workflow integration.

Nanocluster Catalysis: Reactivity and catalytic properties of aluminum nanoclusters: dinitrogen activation, C–I bond dissociation (CC-coupling), site selectivity via reactivity descriptors; DFT and AIMD of transition-metal-doped clusters.

HPC & Computing: IBM Quantum Cloud; massively parallel CPU/GPU architectures; Linux/HPC environments; Snakemake automation.

Education

CSIR–National Chemical Laboratory (NCL)

Pune, Maharashtra, India

Ph.D. in Chemistry

2010–2015

PI: Prof. Sourav Pal. Thesis: “Theoretical Investigation on Structure and Reactivity Properties of Molecules and Metal Clusters: A Conceptual DFT and *Ab initio* Molecular Dynamics Approach.” Funded by the CSIR Research Fellowship (Junior Research Fellow, All India Rank 46; promoted to Senior Research Fellow). **Ph.D. in Chemistry awarded August 2015.**

Visva-Bharati (A Central University)*M.Sc. in Chemistry*

Two-year postgraduate degree. Specialisation: Physical Chemistry (Special Paper); Organic and Inorganic Chemistry.

M.Sc. degree awarded 2009.**West Bengal, India**

2007–2009

University of Burdwan*B.Sc. (Chemistry Honours)*Three-year undergraduate degree. Subjects: Chemistry (Honours), Physics, Mathematics. **B.Sc. degree awarded 2007.****West Bengal, India**

2004–2007

Professional Experience

Cleveland Clinic Foundation (CCF), Cleveland Clinic Research Cleveland, Ohio, United States*Research Associate**Oct 2023–present*

PI: Prof. Kenneth M. Merz Jr., Computational Life Sciences (CLS). Research spans: (i) quantum computing for electronic structure and molecular energetics — development and execution of the EWF–SQD pipeline for FLiBe (LiF–BeF₂) molten salt clusters on IBM quantum hardware; quantum computing for molecular energetics (IBM Quantum, Qiskit); (ii) QM/ML methods for metabolomics and precision medicine.

Michigan State University, Department of Chemistry East Lansing, Michigan, United States*Postdoctoral Fellow**Apr 2019–Oct 2023*

PI: Prof. Kenneth M. Merz Jr. QM/ML spectroscopy workflows for metabolomics (NMR chemical shift and CCS prediction); AutoGraph conformational clustering algorithm; web server development for metabolomics characterization (pomics.org).

Bar-Ilan University, Department of Chemistry*Postdoctoral Research Fellow***Ramat-Gan, Israel***Aug 2015–Apr 2019*

PI: Prof. Dan Thomas Major. Hybrid QM/MM simulation of enzyme-catalysed terpene biosynthesis (selinadiene synthase, limonene synthase); development of CHARMM-based QM/MM docking code (EnzyDock); homology modelling (MODELLER); QM region size convergence in QM/MM; photovoltaic applications (DFT of Te/Y-doped NbBiO₄).

Research Experience

Quantum Computing & EWF–SQD: Designed and executed the EWF–SQD quantum hardware workflow for electronic structure benchmarking of FLiBe (LiF–BeF₂) molten salt clusters. High-entanglement fragments solved on IBM QPUs; democratic partitioning of reduced density matrices; achieves ≤ 0.7 kcal/mol accuracy relative to classical FCI. Demonstrated near-term quantum hardware viability for strongly correlated, energy-relevant ionic systems.

QM/MM & Enzyme Catalysis: Developed QM/MM simulation and docking tools including EnzyDock (CHARMM-based, incorporating simulated annealing MD and Monte Carlo sampling) for studying enzyme catalysis: terpene synthases, Diels–Alder reactions, racemases, and covalent docking. QM region size convergence studies in DNA proton transfer systems.

QM + ML Spectroscopy: Created QM/ML pipelines for NMR chemical shift and Collisional Cross Section (CCS) prediction to enable metabolite structure elucidation; AutoGraph autonomous graph-based conformational clustering; web server development for metabolomics characterization (pomics.org); integrated with Snakemake workflow management.

Nanocluster Chemistry: Applied DFT, *ab initio* MD, and Coupled-Cluster methods to aluminum nanocluster catalysis: N₂ activation by Si/P-doped clusters, oxidative addition of C–I bonds, transition-metal-doped cluster spin states; water cluster benchmarking; Conceptual DFT reactivity descriptors.

Biomolecule Modelling: Homology modelling of proteins (MODELLER); QM/MM simulation of biomacromolecules in physiological environments; computer-aided drug design and protein engineering; protein–ligand docking (EnzyDock, Glide, CDOCKER).

28. **Susanta Das**, Subhamoy Bhowmik, Zhen Li, Milana Bazayeva, Danil Kaliakin, Akhil Shajan, Kenneth M. Merz Jr. *Quantum Computing Enabled ab initio Molecular Dynamics Simulations*. **arXiv preprint arXiv:2607.28548 (2026)**.
doi:10.48550/arXiv.2607.28548
27. **Susanta Das**, Thiago J. Pinheiro Dos Santos, Subhamoy Bhowmik, Milana Bazayeva, Zhen Li, Akhil Shajan, Danil Kaliakin, Fangchun Liang, Vyacheslav S. Bryantsev, Al Geist, Abigail McClain Gomez, Thaddeus Pellegrini, Robert Walkup, Seetharami R. Seelam, Mario Motta, Kenneth M. Merz Jr., Thomas Beck. *Quantum Computations on Fusion Blanket Molten Salts*. **arXiv preprint arXiv:2606.30402 (2026)**.
doi:10.48550/arXiv.2606.30402
26. Zhen Li, Milana Bazayeva, Thaddeus Pellegrini, Mario Motta, Subhamoy Bhowmik, **Susanta Das**, Danil Kaliakin, Fangchun Liang, Akhil Shajan, and Kenneth M. Merz Jr. *Protein-Ligand Free Energy Perturbation on Quantum Hardware*. **arXiv preprint arXiv:2604.09857 (2026)**.
doi:10.48550/arXiv.2604.09857
25. Akhil Shajan, Danil Kaliakin, Fangchun Liang, Thaddeus Pellegrini, Hakan Doga, Subhamoy Bhowmik, **Susanta Das**, Antonio Mezzacapo, Mario Motta, and Kenneth M. Merz Jr. *Molecular Quantum Computations on a Protein*. **J. Chem. Theory Comput.** **2026**, 22, 12, 6041–6056.
doi:10.1021/acs.jctc.6c00364
24. **Susanta Das** and Kenneth M. Merz Jr. *Exploring the Frontiers of Computational NMR: Methods, Applications, and Challenges*. **Chemical Reviews**, **2025**, 125, 19, 9256.
doi:10.1021/acs.chemrev.5c00259
23. Milana Bazayeva, Zhen Li, Danil Kaliakin, Fangchun Liang, Akhil Shajan, **Susanta Das**, and Kenneth M. Merz Jr. *Quantum-Centric Alchemical Free Energy Calculations*. **J. Chem. Theory Comput.** **2026**, 22, 13, 6835–6843.
doi:10.1021/acs.jctc.6c00526
22. Akhil Shajan, Danil Kaliakin, Abhishek Mitra, Javier Robledo Moreno, Zhen Li, Mario Motta, Caleb Johnson, Abdullah Ash Saki, **Susanta Das**, Iskandar Sitdikov, Antonio Mezzacapo and Kenneth M. Merz Jr. *Toward Quantum-Centric Simulations of Extended Molecules: Sample-Based Quantum Diagonalization Enhanced with Density Matrix Embedding Theory*. **J. Chem. Theory Comput.** **2025**, 21, 14, 6801.
doi:10.1021/acs.jctc.5c00114
21. Danil Kaliakin, Akhil Shajan, Javier Robledo Moreno, Zhen Li, Abhishek Mitra, Mario Motta, Caleb Johnson, Abdullah Ash Saki, **Susanta Das**, Iskandar Sitdikov, Antonio Mezzacapo and Kenneth M. Merz Jr. *Accurate quantum-centric simulations of intermolecular interactions*. **Nature Commun. Phys.** **2025**, 8, 396.
doi:10.1038/s42005-025-02305-9
20. **Susanta Das** and Kenneth M. Merz Jr. *Molecular Gas-Phase Conformational Ensembles*. **J. Chem. Inf. Model.** **2024**, 64, 749.
doi:10.1021/acs.jcim.3c01309
19. **Susanta Das**, Laleh Dinpazhoh, Kiyoto Aramis Tanemura, Kenneth M. Merz Jr. *Rapid and Automated Ab Initio Metabolite Collisional Cross Section Prediction from SMILES Input*. **J. Chem. Inf. Model.**

2023, 63, 4995.

doi:10.1021/acs.jcim.3c00890

18. **Susanta Das**, Kiyoto Aramis Tanemura, Laleh Dinpazhoh, Mithony Keng, Christina Schumm, Lydia Leahy, Carter K. Asef, Markace Rainey, Arthur S. Edison, Facundo M. Fernández, Kenneth M. Merz Jr. *In Silico Collision Cross Section Calculations to Aid Metabolite Annotation*. **J. Am. Soc. Mass Spectrom.** **2022**, 33, 5, 750.
doi:10.1021/jasms.1c00315
17. Ricardo M. Borges, Sean M. Colby, **Susanta Das**, Arthur S. Edison, Oliver Fiehn, Tobias Kind, Jesi Lee, Amy T. Merrill, Kenneth M. Merz Jr., Thomas O. Metz, Jamie R. Nunez, Dean J. Tantillo, Lee-Ping Wang, Shunyang Wang, and Ryan S. Renslow. *Quantum Chemistry Calculations for Metabolomics*. **Chem. Rev.** **2021**, 121, 10, 5633.
doi:10.1021/acs.chemrev.0c00901
16. Kiyoto Aramis Tanemura, **Susanta Das**, and Kenneth M. Merz Jr. *AutoGraph: Autonomous Graph-Based Clustering of Small-Molecule Conformations*. **J. Chem. Inf. Model.** **2021**, 61, 4, 1647.
doi:10.1021/acs.jcim.0c01492
15. **Susanta Das**, Arthur S. Edison, and Kenneth M. Merz Jr. *Metabolite Structure Assignment Using In Silico NMR Techniques*. **Anal. Chem.** **2020**, 92, 15, 10412.
doi:10.1021/acs.analchem.0c00768
14. **Susanta Das**, Mor Shimshi, Keren Raz, Neta Nitoker Eliaz, Anil Ranu Mhashal, Tamar Ansbacher, and Dan T. Major. *EnzyDock: Protein–Ligand Docking of Multiple Reactive States along a Reaction Coordinate in Enzymes*. **J. Chem. Theory Comput.** **2019**, 15, 9, 5116.
doi:10.1021/acs.jctc.9b00366
13. **Susanta Das**, Kwangho Nam, and Dan Thomas Major. *Rapid Convergence of Energy and Free Energy Profiles with Quantum Mechanical Size in Quantum Mechanical–Molecular Mechanical Simulations of Proton Transfer in DNA*. **J. Chem. Theory Comput.** **2018**, 14, 1695.
doi:10.1021/acs.jctc.7b00964
12. (Book Chapter) Dan T. Major, Reuven Eitan, **Susanta Das**, Anil R. Mhashal, Vijay Singh. *Nuclear Quantum Effects in Enzymatic Reactions*. In: *Simulating Enzyme Reactivity: Computational Methods in Enzyme Catalysis*, Eds. Inaki Tunon and Vicent Moliner. **The Royal Society of Chemistry**, **2017**, Chapter 9, pp. 340–376.
doi:10.1039/9781782626831-00340
11. (Book Chapter) Mudit Dixit, **Susanta Das**, Anil R. Mhashal, Reuven Eitan, Dan T. Major. *Practical Aspects of Multiscale Classical and Quantum Simulations of Enzyme Reactions*. In: *Computational Approaches for Studying Enzyme Mechanism*, Ed. Gregory A. Voth. **Methods in Enzymology**, **2016**, 577, pp. 251–286.
doi:10.1016/bs.mie.2016.05.046
10. **Susanta Das**, Mudit Dixit, Dan T. Major. *First Principles Model Calculations of the Biosynthetic Pathway in Selinadiene Synthase*. **Bioorg. Med. Chem.** **2016**, 24, 4867.
doi:10.1016/j.bmc.2016.07.002
9. Bosmat L. Hevroni, Dan T. Major, Mudit Dixit, Anil R. Mhashal, **Susanta Das**, Bilha Fischer. *Nucleoside-2',3'/3',5'-bis(thio)phosphate Antioxidants Are Also Capable of Disassembly of Amyloid β_{42} -Zn(ii)/Cu(ii) Aggregates via Zn(ii)/Cu(ii)-Chelation*. **Org. Biomol. Chem.** **2016**, 14, 4640.
doi:10.1039/C6OB00613B

8. Turbasu Sengupta, **Susanta Das**, Sourav Pal. *Transition Metal Doped Aluminum Clusters: An Account of Spin*. **J. Phys. Chem. C**, **2016**, 120, 10027.
doi:10.1021/acs.jpcc.6b00510
7. **Susanta Das**, Turbasu Sengupta, Achintya K. Dutta, Sourav Pal. *Electron Detachment and Subsequent Structural Changes of Water Clusters*. **J. Phys. Chem. A**, **2016**, 120, 1065.
doi:10.1021/acs.jpca.5b09389
6. Rupesh L. Gawade, Debamitra K. Chakravarty, Amol Kotmale, Ekta Sangtani, Pranaya V. Joshi, Awais Ahmed, Manoj V. Mane, **Susanta Das**, Kumar Vanka, Pattuparambil R. Rajamohanan, Vedavati G. Puranik, Rajesh G. Gonnade. *Additive Mediated Syn–Anti Conformational Tuning at Nucleation to Capture Elusive Polymorphs: Remarkable Role of Extended π -Stacking Interactions in Driving the Self-Assembly*. **Cryst. Growth Des.** **2016**, 16, 2416.
doi:10.1021/acs.cgd.6b00204
5. Turbasu Sengupta, **Susanta Das**, Sourav Pal. *Oxidative Addition of the C–I Bond on Aluminum Nanoclusters*. **Nanoscale**, **2015**, 7, 12109.
doi:10.1039/C5NR02278A
4. **Susanta Das**, Sailaja Krishnamurty, and Sourav Pal. *Dinitrogen Activation by Silicon and Phosphorus Doped Aluminum Clusters*. **J. Phys. Chem. C**, **2014**, 118, 19869.
doi:10.1021/jp505700a
3. **Susanta Das**, Sailaja Krishnamurty and Sourav Pal. *Understanding the Site Selectivity in Small-Sized Neutral and Charged Al_n ($4 \leq n \leq 7$) Clusters Using Density Functional Theory Based Reactivity Descriptors: A Validation Study on Water Molecule Adsorption*. **J. Phys. Chem. A**, **2013**, 117, 8691.
doi:10.1021/jp403109s
2. **Susanta Das**, Sapana V. Shedge and Sourav Pal. *Critical Study of the Charge Transfer Parameter for the Calculation of Interaction Energy Using the Local Hard–Soft Acid–Base Principle*. **J. Phys. Chem. A**, **2013**, 117, 10933.
doi:10.1021/jp407070h
1. Deepti Mishra, **Susanta Das**, Sailaja Krishnamurty and Sourav Pal. *Understanding the Orientation of Water Molecules Around the Phosphate and Attached Functional Groups in a Phospholipid Molecule: A DFT-Based Study*. **Molecular Simulation**, **2013**, 39, 937.
doi:10.1080/08927022.2013.783701

Technical Skills / Expertise

Quantum Computing: IBM Quantum Platform; Qiskit; Sample-based Quantum Diagonalization (SQD); EWF–SQD pipeline; quantum hardware benchmarking; reduced density matrix methods.

Workflow: Snakemake (workflow management); Python-based automation; HPC job scheduling.

QM Software: GAUSSIAN, QUICK, GAMESS, QCHEM, TURBOMOLE, DEMON-2K, VASP, MOLPRO.

MM Software: CHARMM, AMBER, NAMD.

QM/MM: CHARMM–QCHEM interface; hybrid QM/MM setup and analysis.

Method Dev.: EnzyDock (enzyme QM/MM docking); NMR Chemical Shift prediction workflow; Collisional Cross Section (CCS) prediction workflow; AutoGraph conformational clustering; EWF–SQD pipeline for ionic materials.

Docking: EnzyDock (in-house); Glide (Schrödinger Maestro); CDocker.

Programming: Fortran, Python, Shell Scripting (Bash/Zsh).

Modelling & Viz.: PyMOL, VMD, CHIMERA, AVOGADRO, MOLDEN, Maestro (Schrödinger), Discovery Studio, MODELLER, GNUPLOT, ChemDraw, VESTA, P4VASP.

HPC / Facilities: IBM Quantum Cloud; CPU/GPU massively parallel architectures; Linux/HPC.

Professional: Grant writing; peer-reviewed manuscript preparation; international collaborative research; technical mentoring; HPC system administration.

Scientific Presentations and Conference Participation

Oral Presentations:

08/2026: Texas Quantum Summit 2026 — panel: *Quantum-centric Supercomputing for Healthcare and Life Sciences*, Breakout Session 3, Davidson Gundy Alumni Center, University of Texas at Dallas, Richardson, TX, USA (August 13–14, 2026). Panel led by Nick Brønne (IBM Quantum). *[Invited panelist]*

04/2026: IBM Quantum Ideation Workshop — *Molecular Simulation at the Quantum Frontier*, IBM Quantum Computational Sciences Program, Building 60 Main Auditorium (April 15–16, 2026). Oral presentation: *Protonation State Sampling via IBM Quantum QCSC*.

09/2020: MANA Metabolomics Association of North America, University of Michigan, Ann Arbor, USA.

11/2019: MANA Metabolomics Association of North America, University of Georgia, Atlanta, USA.

Conferences, Symposiums, and Workshops Attended:

04/2024: Healthcare and Life Sciences Quantum Working Group Meeting (HCLS), Cleveland, OH, USA.

09/2022: MANA Metabolomics Association of North America, University of Alberta, Edmonton, Canada.

10/2021: MANA Metabolomics Association of North America, Ohio State University, Columbus, OH, USA.

05/2018: 1st French-Israeli Workshop on Cheminformatics, Bar-Ilan University, Ramat-Gan, Israel.

02/2018: 83rd Annual Meeting of The Israel Chemical Society, Tel-Aviv, Israel.

02/2017: 82nd Annual Meeting of The Israel Chemical Society, Tel-Aviv, Israel.

02/2017: Molcas Users Workshop, Hebrew University of Jerusalem, Jerusalem, Israel.

02/2016: Workshop on High Throughput Materials Discovery: Perspectives and Challenges in Theory and Experiment, Tel-Aviv University, Tel-Aviv, Israel.

01/2016: 23rd Jerusalem School in Life Sciences: Computational and Structural Biology, Hebrew University of Jerusalem, Jerusalem, Israel.

06/2014: Summer School MASTANI — Materials Simulations: Theory and Numerics, IISER Pune, Pune, India.

09/2013: International Conference on Current Trends in Theoretical Chemistry (CTTC-VI), Jagiellonian University, Krakow, Poland. *[Poster presented]*

02/2013: ESDMC — International Conference on Electronic Structure and Dynamics of Molecules and Clusters, IACS (Indian Association for the Cultivation of Science), Kolkata, India. *[Poster presented]*

12/2012: TCS — Theoretical Chemistry Symposium, IIT Guwahati (IITG), Guwahati, India. *[Poster presented]*

Reviewer Activities

41 fully documented peer-review assignments across 14 journals (2017–2026); verified via invitation, acceptance, and acknowledgement correspondence.

2017: Nanoscale (Royal Society of Chemistry).

2019: Journal of Molecular Liquids (Elsevier).

2022: Structural Chemistry (Springer); Theoretical Chemistry Accounts (Springer); Journal of Chemical Information and Modeling (ACS).

2023: Journal of Chemical Information and Modeling (ACS, multiple papers); ACS Omega (ACS); Journal of Computer-Aided Molecular Design (Springer); MDPI-Coatings; MDPI-Crystals; MDPI-Electronics; MDPI-International Journal of Molecular Sciences; MDPI-Materials; MDPI-Molecules (multiple papers); MDPI-Sustainability; MDPI-Symmetry (multiple papers).

2024: Journal of Chemical Information and Modeling (ACS); ACS Omega (ACS); MDPI-Ceramics; MDPI-Crystals; MDPI-Journal of Composites Science; MDPI-Materials.

2025: Journal of Chemical Information and Modeling (ACS, multiple papers); ACS Omega (ACS); Journal of Organic Chemistry (ACS); MDPI-International Journal of Molecular Sciences; MDPI-Molecules (multiple papers); MDPI-Processes.

2026: Journal of Chemical Information and Modeling (ACS).

Honors and Awards

2009: CSIR–UGC National Eligibility Test (NET) — Junior Research Fellowship (JRF) awarded; **All India Rank 46** (among all Indian universities). Qualified **Graduate Aptitude Test in Engineering (GATE)**, 2009. These national competitive examinations are the primary route to government-funded doctoral research fellowships in India.